

Lumina: Confidence-Weighted Identity Fusion and Preference-Aware Ranking for Photo Curation

Agustinus Benyamin Prasetyo and Ong Qing Quan
Institute of Systems Science, National University of Singapore
Singapore 119615
Capstone Group 10

Abstract—Lumina curates a personal photo batch into events and people, then selects a best shot of each person at each event. The Semester 1 prototype already combined face embeddings, body re-identification, scene embeddings and a weighted aesthetic ranker, but identity was not fused, event boundaries depended on who appeared in the frame, and the ranker could not express a blink, a near-duplicate burst, or a user’s taste. This paper specifies the capstone model. Faces are matched to person boxes by a one-to-one assignment with a head-position prior. ArcFace and OSNet embeddings are placed in separate blocks of a concatenated vector, scaled by a confidence weight $\alpha \in [0.7, 1]$, so that cosine similarity in the fused space is an α -weighted blend of the two modalities. Events are agglomerative clusters on DINOv3 distances blended with capture time, named by CLIP against a closed prompt bank. Ranking uses seven signals with floored normalisation, an absolute eyes-open score, and a hard constraint that a detected blink loses to any eyes-open alternative. A Bradley–Terry update turns a user’s swap into one gradient step on the weight vector. On a private set of 55 photographs in 20 occasions, the label-free event threshold reaches an adjusted Rand index of 0.985 against the folder grouping, against 0.549 for the Semester 1 event features, and identity clustering makes none of the 526 merges that would put two faces from one photograph in the same person. Thirteen people choosing which frame to keep score the displayed order at 0.53 against 0.50 for a random pick: above chance, and too small a margin to keep only one frame or to delete another without a stated reason. Preferences vary, within and across people. On one occasion every tester kept the frame that shows the whole party, which the ranker had placed last. One preference pass does not raise held-out agreement.

Index Terms—photo curation, face recognition, person re-identification, event clustering, learning to rank, preference learning

I. Introduction

A holiday, a wedding or a family weekend produces a pile of near-duplicate photographs. The useful output is not another grid of files. It is a small set of events, a cast of people, and one defensible shot of each person in each event. That task has three coupled decisions: who is the same person when the face is small, turned or missing; which photographs belong to the same occasion when the background barely changes; and which frame is the one to keep when several are acceptable.

Phone galleries cluster faces. They do not rank a burst, explain the choice, or change that choice when the user prefers another frame. Lumina composes published

encoders and puts the modelling effort into the fusion, the clustering objective, the constraints on the ranker, and a preference update from a handful of pairwise edits.

The Semester 1 baseline detected people with YOLOv8 [5], embedded bodies with OSNet [4], clustered them with HDBSCAN [6], embedded scenes with a DINO-family model, scored aesthetics with NIMA [7], and ranked each event with seven hand-set weights. Its own measurements, on a few dozen photographs, already pointed at the next model rather than at more of the same pipeline. HDBSCAN’s silhouette on re-identification embeddings was about 0.25, against about 0.79 for a tuned DBSCAN. Adding a person co-attendance block to the event features raised silhouette only from 0.764 to 0.775. An ablation of the ranker found that the eye-aspect term never changed a winner, which is evidence that the term had no usable dynamic range, not evidence that blinks are unimportant.

This paper treats those findings as the problem statement. The contributions are:

- 1) a confidence-weighted concatenation of ArcFace [2] and OSNet that keeps cosine similarity equal to a modality blend, with a floor on the face weight so that a face with a body and the same face without one still meet;
- 2) event clustering on time-blended DINOv3 distances, dropping identity from the event features so that a clustering error about people cannot define the occasion;
- 3) a ranker whose normalisation, eyes-open score and blink constraint are specified so that burst noise and closed eyes cannot win on a rescaled aesthetic gap;
- 4) an online Bradley–Terry update [8] of the seven weights, with the operating step scored by asking people which photograph they would keep;
- 5) an identity check, using the same ArcFace model, that discards a generative retouch when the weakest face match falls below a set cosine.

Items 1–4 are what Section V measures. Naming, search, captions and retouching (Section IV) read that output and do not change it. There is no public-album benchmark.

II. Related work

Time has been an event feature since PhotoTOC, which clustered photographs by creation time and colour [1].

Later galleries replaced colour with learned scene descriptors and added a separate face clustering. Lumina keeps that split and adds a ranker between them.

ArcFace embeddings are compared by cosine [2]; the detector and the 512-dimensional face vector are InsightFace buffalo_1 [3]. OSNet supplies a 512-dimensional body vector when the face is missing [4], from YOLOv8 boxes [5]. The two vectors are not one space, so they are concatenated rather than averaged. HDBSCAN fits a density that has no known threshold [6]; a face embedding trained for a cosine cutoff does not. Scene descriptors are DINOv3, with DINOv2 as a fallback [9], [10]. CLIP names an event from a fixed prompt list [11]. BLIP captions the selected frame only [12]. NIMA contributes one aesthetic term, not the rank [7], [13]. MMR trades that score against redundancy when the output is a set [14]. Bradley–Terry updates the weights from a pairwise swap [8]. Silhouette picks the event threshold and is not a substitute for a label [15].

III. Model

A job is a set of images $\{I_i\}_{i=1}^N$, N up to a few hundred. Each image may carry an EXIF capture time t_i . The model returns a partition of images into events, a set of identity clusters, a total order inside each event, a best photograph per person inside each event, three highlight sets, optional text-query ranks, and a weight vector that later jobs can reuse. The application shows events as moments. Figure 1 is the data flow. Encoders are frozen. Everything after the embeddings is deterministic given the configuration in Table I. Association, fusion, events, the ranker and preference learning decide the curation. Naming, search, captions and retouching do not feed back into it.

A. Associating a face with a body

Semester 1 linked a face to the first person box that contained it. Overlapping people in a group photograph then shared a face. The replacement is a greedy one-to-one assignment. For face box f and person box p , let $c(f, p)$ be the fraction of the face area inside the person box. Pairs with $c(f, p) < 0.5$ are discarded. Among the rest the score is

$$c(f, p) + 0.5 h(f, p), \quad (1)$$

where the head prior h is high when the face centre is near the horizontal centre of the person box and in its upper portion. With person width w and height h_p , and face centre (x, y) ,

$$h = \max(0, 1 - \Delta x) \max(0, 1 - \min(\Delta y / 0.6, 1)), \quad (2)$$

$\Delta x = |x - \bar{x}_p|/w$ and $\Delta y = (y - y_p^{\text{top}})/h_p$. Candidates are taken in descending score. Each face and each person box is used at most once. A body with no assigned face, and a face with no assigned body, both remain in the pipeline; fusion treats the missing block as empty rather than imputing it.

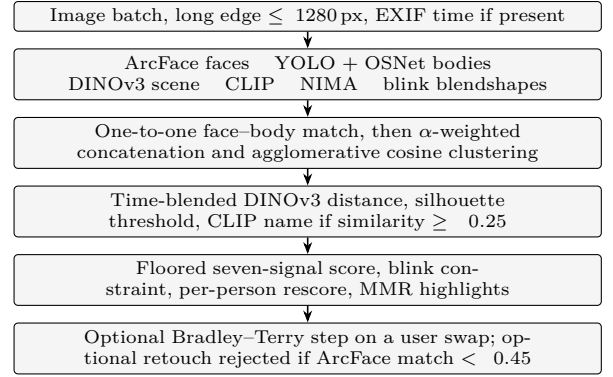


Fig. 1. Capstone inference path. Encoders are pretrained and frozen. Identity and events are separate partitions. Preference learning and retouching run only when the user asks.

TABLE I
Operating points used by the capstone model.

Choice	Value
Face embedding	InsightFace buffalo_1, 512-d
Body embedding	OSNet-x1.0, 256 × 128, 512-d
Scene embedding	DINOv3 ViT-S/16 (DINOv2-base fallback)
Text–image	CLIP ViT-B/32
Aesthetic	NIMA (AVA), one scalar in [0, 10]
Face cluster distance	0.60 cosine (face-only ablation)
Fused cluster distance	0.58 cosine
Body attach distance	0.75 Euclidean to ReID centroid
α range	[0.7, 1.0]
Event time weight, τ	0.35, 6 hours
Event threshold sweep	[0.05, 1.0] step 0.025, patience 10
Duplicate cosine	≥ 0.965 on DINOv3
Blink reject	eyes-open < 0.45 and a face was detected
MMR λ	0.90 / 0.70 / 0.45 (at most 6)
Preference step, prior pull	$\eta = 0.20$, $\lambda_{\text{prior}} = 0.05$
Retouch reject / warn	ArcFace cosine 0.45 / 0.65

B. Confidence-weighted embedding fusion

Let $\hat{f}$ and $\hat{b}$ be L2-normalised ArcFace and OSNet vectors. They are not added. The fused descriptor is the concatenation

$$u = \begin{bmatrix} \sqrt{\alpha} \hat{f} \\ \sqrt{1 - \alpha} \hat{b} \end{bmatrix}, \quad (3)$$

which already has unit norm when both blocks are present. For two detections with weights α_i and α_j ,

$$u_i^T u_j = \sqrt{\alpha_i \alpha_j} \hat{f}_i^T \hat{f}_j + \sqrt{(1 - \alpha_i)(1 - \alpha_j)} \hat{b}_i^T \hat{b}_j. \quad (4)$$

When $\alpha_i = \alpha_j = \alpha$ this is exactly α times the face cosine plus $(1 - \alpha)$ times the body cosine. A missing modality forces α to 1 (face only) or 0 (body only) and zeros the other block, so similarity falls back to the modality the two detections share.

The weight itself is a detection-quality score, not a learned gate:

$$\gamma = \text{clip}\left(\frac{s_{\text{det}} - 0.35}{0.85 - 0.35}, 0, 1\right), \quad (5)$$

$$\sigma = \text{clip}(r_{\text{face}}/0.02, 0, 1), \quad (6)$$

$$\alpha = 0.7 + 0.3\gamma(0.5 + 0.5\sigma). \quad (7)$$

Here s_{det} is the InsightFace detection score and r_{face} is the face area as a fraction of the image. A large, confident face gets $\alpha = 1$. A weak or tiny face gets α near 0.7, never below, so that a face matched to a body and the same face alone still meet under (4). At a face cosine of 0.50, the $\alpha = 0.7$ scaling puts the fused distance near 0.58, which is the agglomerative threshold. The face-only ablation uses 0.60.

Clustering is average-linkage agglomerative clustering on cosine distance [16]. HDBSCAN on OSNet, with noise points reassigned to the nearest centroid, is used only when the batch has no usable face embedding, and as the body_only ablation. After face clusters exist, a person box that still has no identity is attached to the nearest ReID centroid if the Euclidean distance is below 0.75.

A final filter decides which clusters are subjects rather than background. Public scenes produce a singleton cluster for every bystander. A cluster is shown if it appears in at least two photographs or if its largest face covers at least 0.004 of some image (a square face of about 80 pixels on the 1280-pixel working resolution). Other clusters stay in the index and do not enter the cast.

C. Events from scene and time

Semester 1 appended a person co-attendance block to the scene vector. The silhouette gain was small, and an identity error rewrote the event feature of every frame that person appeared in. The capstone event feature drops that block.

Let rows of X be L2-normalised. Visual distance is $D_{ij}^{\text{vis}} = \max(0, 1 - x_i^T x_j)$. Where both photographs have EXIF time,

$$D_{ij} = (1 - w) D_{ij}^{\text{vis}} + w \min(|t_i - t_j|/\tau, 1), \quad (8)$$

with $w = 0.35$ and $\tau = 6$ hours. A missing timestamp leaves the pair at its visual distance, so a batch stripped of EXIF degrades to pure scene clustering rather than failing. Average-linkage agglomerative clustering is run on D across distance thresholds from 0.05 to 1.0 in steps of 0.025. The threshold with the best silhouette on the precomputed distance [15] is kept, and the sweep stops after ten consecutive thresholds that do not improve. The earlier sweep, from 0.1 to 3.0 in steps of 0.25 with patience 3, is the one Section V-C shows landing on the wrong side of the silhouette peak. Blended distances already lie near $[0, 1]$, so extending the grid past 1 added nothing. Two identical scenes taken more than τ apart have blended distance 0.35, so time can separate them only when the

chosen threshold is below that. A wrong clock does not override a clearly different scene: the visual term still carries weight 0.65.

D. The seven-signal ranker

Inside an event, each photograph receives a composite score $S = \sum_{k=1}^7 w_k z_k$ with $\sum_k w_k = 1$ and z_k a normalised signal. Table II lists the signals and the capstone weights. Two weights differ from Semester 1: eye openness moves from 0.05 to 0.10, and detection confidence from 0.10 to 0.05. Detection confidence measures whether a box is reliable. It is a weak statement about whether the photograph is the one to keep. Eye openness is the opposite, once it is measured on an absolute scale.

Plain min-max normalisation is the wrong map for a burst. If every NIMA score in the event lies between 5.00 and 5.05, min-max stretches that gap to 0 versus 1 and the aesthetic term dominates the rank for no visual reason. Floored min-max divides by $\max(\max z - \min z, c_k)$ with the floors in Table II. A 0.05-point NIMA gap then stays near $0.05/0.75 \approx 0.07$ instead of 1. Centrality uses the same idea on Euclidean distance to the mean embedding. The mean of unit vectors is not itself unit, so this is distance-to-centroid, not cosine-to-a-second normalisation. The Semester 1 write-up described centrality as cosine to the centroid; the implementation now matches the floored Euclidean form above.

Pose is

$$q = \text{clip}\left(1 - 0.5\frac{|\text{yaw}|}{90} - 0.3\frac{|\text{pitch}|}{90} - 0.2\frac{|\text{roll}|}{45}, 0, 1\right), \quad (9)$$

with angles in degrees. Landmark eye-aspect ratio does not separate open from closed eyes on this detector [17], so eyes-open uses MediaPipe blink blendshapes [18]: $1 - \max(\text{blink}_L, \text{blink}_R)$. If the landmarker misses the face, the fallback is $\text{clip}((\text{EAR} - 0.15)/0.10, 0, 1)$, which maps a blink to 0 and a clearly open eye to 1 without looking at the rest of the event. In a group photograph the photo-level eyes-open score is the minimum across faces, so one blinking person disqualifies the frame. Sharpness, pose and detection confidence are averaged; face size takes the maximum.

The score alone still lets a technically sharp, high-NIMA blink win, which is the failure that the weight change does not fix. After scoring, any photograph with a detected face and eyes-open below 0.45 is ordered after every non-blinking photograph. Ties inside each block break by S . If every frame is a blink, the constraint does not empty the event: the best blink remains the selection, and it is not flagged as a reject. The same rule is applied when each person's own photographs are re-scored. For that per-person order, centrality and NIMA stay at their event-level values, and the five face signals are recomputed on that person's faces. Eyes stay absolute. The other four face signals are ordinary min-max across that person's frames in the event; the floors of Table II are not reapplied there.

TABLE II

Ranker weights. Semester 1 weights in parentheses where they differ. Centrality and the five face signals other than eyes use floored min-max inside the event. Eyes-open is absolute.

Signal	Weight	What is measured
Centrality	0.25	1 minus floored min-max of $\ x - \bar{x}\ _2$ to the event's DINOv3 mean. Floor span 0.06.
NIMA	0.25	Aesthetic mean in $[0, 10]$. Floor span 0.75 points.
Sharpness	0.15	Laplacian variance of the face region. Floor span 60.
Face size	0.10	Face area over image area. Floor span 0.008.
Detection	0.05 (0.10)	InsightFace score. Floor span 0.10.
Pose	0.10	Frontal score from yaw, pitch and roll, Eq. (9). Floor span 0.25.
Eyes	0.10 (0.05)	Eyes-open in $[0, 1]$, not rescaled.

A very small sharpness gap can therefore still dominate a one-person burst. That inconsistency is recorded in Section VI.

Near-duplicates are the connected components of DINOv3 pairs with cosine at least 0.965. The highest-ranked member of a component is the keeper. The others are marked as duplicates of it. A non-winner is also flagged when the face Laplacian variance is below 25, or when eyes-open is below 0.45. Photographs with no face are not flagged for blur or blink. The selected photograph of the event is never flagged. A deletion reason is a template over the same normalised signals. It can name the term the score used. It cannot invent a reason the score did not have.

MMR builds a set of at most six highlights,

$$i_{t+1} = \arg \max_{i \notin S_t} \left[\lambda S_i - (1 - \lambda) \max_{j \in S_t} \cos(x_i, x_j) \right], \quad (10)$$

with x the DINOv3 embedding and $\lambda \in \{0.90, 0.70, 0.45\}$ for a quality-leaning, balanced, or diverse set [14]. The first pick is $\arg \max_i S_i$. Because the blink constraint reorders the best-shot list without changing S_i , that first highlight can be a blink the best-shot rule rejected. The two outputs are not the same policy.

E. Preference learning

A swap — the user promotes photograph B over the system's choice A — is one Bradley–Terry observation. With normalised signal vectors s_A and s_B and weights w on the simplex,

$$P(B \succ A) = \sigma(w^\top (s_B - s_A)), \quad \sigma(u) = (1 + e^{-u})^{-1}. \quad (11)$$

One gradient step on the log-likelihood, with a quadratic pull toward the default w_0 of Table II, is

$$w \leftarrow \Pi(w + \eta [(1 - p)(s_B - s_A) - \lambda_{\text{prior}}(w - w_0)]), \quad (12)$$

where p is (11) under the current w , $\eta = 0.20$ and $\lambda_{\text{prior}} = 0.05$. The map Π clips each coordinate at 0.01 and renormalises. The pull toward w_0 is the point: a few

swaps should move the ranker, not replace it with a one-hot fit to noise. Weights persist per user. They do not retrain the encoders.

IV. Supporting components

A. Naming and search

Each event is named by the CLIP centroid against 26 prompts, accepted only if cosine is at least 0.25. Below that floor the name falls back to an EXIF time phrase, or to “Moment k ”. Text search averages four prompt templates and ranks by cosine. An optional call may replace the CLIP label with a short album title, and BLIP may caption the selected frame. Failures keep the CLIP or time label. Neither caption nor title enters clustering or ranking.

B. Identity-guarded retouching

An optional edit is accepted only after every original face is re-matched with ArcFace. The score is the minimum cosine, so one altered face fails the group. Below 0.45 the edit is discarded; from 0.45 to 0.65 it is shown with a warning; at or above 0.65 it is accepted. No detected face means the edit is shown and marked unverified. The cutoffs were not fit on a labelled retouch set. Models that ignore the input photograph are refused, and the original file is never overwritten.

C. Where the model runs

The encoders run in one Python service. Jobs are serial because the loaded networks are not thread-safe. Features are cached by file hash, so a repeated batch skips inference. Photographs leave the service only for the optional retouch and album title. The web application on this service, at <https://lumina-production-639d.up.railway.app/>, also exports an album, a video and a 3D view of the events; none of them feeds back into the model.

V. Evaluation

Each experiment asks whether one decision does what it claims: identity, events, which ranker rule changes the winner, and whether a few human corrections move later choices. Naming, search, captions and retouching are not measured.

A. Dataset and what counts as correct

The set is 55 private photographs grouped into 20 occasions by filename prefix. That grouping is one person's judgement, so event quality is also reported with indices that do not use it. Nobody labelled identities. Thirty-two photographs have an EXIF time. Runs use the service encoders on CPU, including the loaded DINOv3 weights. The harness partition matches the pipeline. A cold pass takes 82 s; a cached repeat takes 1.9 s.

B. Identity, without person labels

Two faces in the same photograph cannot be the same person. Counting how often a clustering does that is a mistake count that needs no labelling, and it is the number we trust most. The second check is stability: drop a random 20% of the detections, cluster again, and measure the adjusted Rand index (ARI) [19] against the clustering of the same detections in the full run, over 30 draws. A clustering that reshuffles when a few photographs are removed is not a reliable cast list. Silhouette is reported only as a description of separation.

Table III is the comparison. Fusing the face and the body, and using the face alone, make none of these impossible merges. Clustering the body alone makes 85 of them, which is the empirical reason the face block carries most of the weight in (3). The production threshold of 0.58 sits inside a band that stays clean: raising it through 0.45, 0.50, 0.55, 0.58 and 0.62 keeps the cannot-link count at zero, and the first violations appear at 0.66 (3 of 526). Stability stays between 0.94 and 0.99 across that sweep, so the threshold is not balanced on one value. Lowering the face-weight floor from 0.7 to 0.5, or raising it to 0.9, also leaves the cannot-link count at zero. The floor is the scaling argument in Section III-B. This set does not contradict it, and it does not prove it.

C. Events against the folder grouping

Event quality uses the folder labels, and we say so wherever a number depends on them. ARI is 1 when the clustering matches the grouping exactly and 0 when it matches no better than chance. Homogeneity is 1 when no cluster mixes two occasions; completeness is 1 when no occasion is split. The threshold itself is chosen by silhouette, which does not see the labels, and the same partition is then scored against them. An oracle row records the best ARI on the same distance matrix if the labels were allowed to pick the threshold. Stability is again an 80% subsample, here of photographs, over 30 draws.

Table IV is the result, and Figure 2 is why the sweep changed. The earlier coarse grid of Section III-C, steps of 0.25 from 0.1, has its silhouette maximum at the first point it tries. That cuts the 20 occasions into 27 clusters (ARI 0.745). Steps of 0.025 put the silhouette peak at 0.175, which is also the threshold the oracle would have chosen, and the ARI rises to 0.985 with 21 clusters. The early stop does not cause it: patience 3, patience 10 and a full sweep of the fine grid all select 0.175 on this matrix. One occasion is still split.

Time is a smaller lever than the grid, but it is not zero. With the fine sweep and no time term the ARI is 0.886; any time weight from 0.2 to 0.5 reaches 0.985, and 0.7 falls back to 0.954. Changing τ from 1 hour to 24 hours, or switching between average, complete and single linkage, leaves the ARI at 0.985. The Semester 1 features, rescored with the same fine sweep, reach only 0.549, and two of their clusters mix photographs taken on different

TABLE III

Identity clustering on 55 photographs. Cannot-link counts pairs of faces in one photograph that were assigned the same person. Stability is the mean ARI over 30 subsamples of 80% of the detections.

Mode	Clusters	Shown	Cannot-link	Stability
Fused (production)	49	42	0 / 526	0.973
Face only	50	41	0 / 526	0.980
Body only	86	83	85 / 976	0.880

TABLE IV

Event clustering. The oracle row uses the folder labels to pick the threshold and is an upper bound, not a method. Semester 1 is scene features plus a person co-attendance block and no time.

Setting	Thr.	k	ARI	Stability
Production	0.175	21	0.985	0.969
Coarse sweep	0.100	27	0.745	0.749
No time, $w=0$	0.225	22	0.886	0.957
Semester 1	0.300	21	0.549	0.874
features				
Oracle, this distance	0.175	21	0.985	—

dates. The production clustering mixes none. Its best ARI at any threshold is 0.600, so the gap is in the features, not in the threshold rule.

D. What moves the best shot

There is no label for the best photograph of an occasion, and this paper does not invent one. The question the set can answer is which part of the ranker changes the choice. Sixteen of the 20 occasions contain at least two photographs. For each variant we count how many of those 16 change their winner relative to the production ranker, the mean Kendall τ between the two orders, and how often the winner is a detected blink when the same occasion also contains an eyes-open frame. NIMA scores on this set run from 3.26 to 4.88.

Table V separates three claims. The blink constraint is not cosmetic: without it, 3 of the 16 winners are closed-eye frames that had an open-eye alternative, and the Semester 1 rule, which also lacks the floors, does the same. An aesthetic score on its own is not a curator. It disagrees with the full ranker on 12 of 16 winners and agrees with the production order worse than a reversed ranking. Centrality and sharpness each move 4 winners; removing NIMA moves 1, and removing face size, detection confidence, pose or the eyes-open weight moves none. The weight can be inert while the constraint is not. Perturbing any single weight by $\pm 20\%$ changes no winner.

E. A human keep-one-photo study

Thirteen people finished a keep-one-photo task on 16 occasions from the same private collection. Each question showed two, three or four photographs from one occasion and asked which single photograph they would keep. Twelve questions updated the ranker: eight, then four more after the held-out block. Two of those occasions were

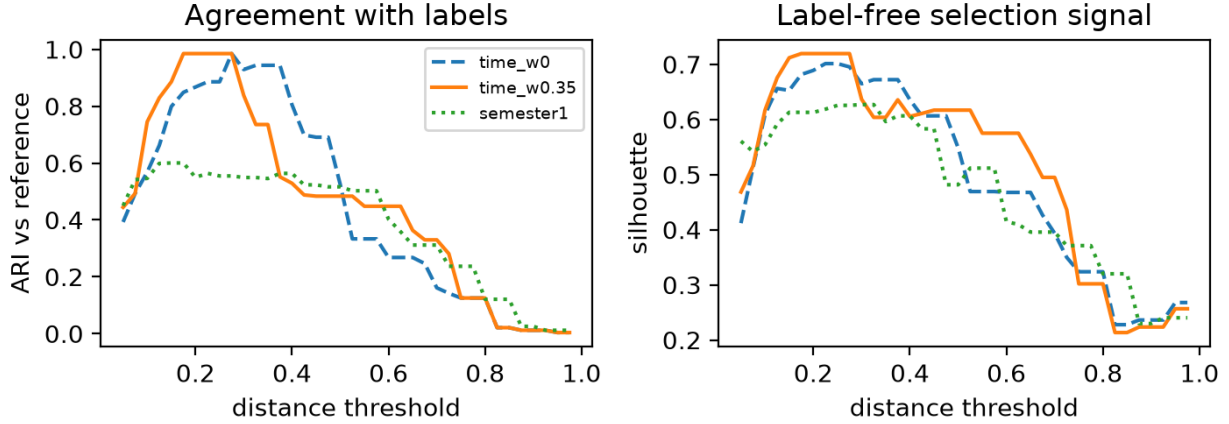


Fig. 2. Event distance threshold on the 55 photographs. Left: agreement with the folder grouping. Right: the silhouette the selection rule actually sees. The production rule takes the silhouette peak of the solid curve.

TABLE V

Best-shot ranker on 16 occasions with at least two photographs. Blink picks are winners with eyes-open below 0.45 in an occasion that had an eyes-open alternative.

Variant	Changed	Kendall τ	Blinks
No span floors	4 / 16	0.66	0
No blink constraint	3 / 16	0.70	3
Semester 1 rule	8 / 16	0.32	3
NIMA alone	12 / 16	-0.15	4
Without centrality	4 / 16	0.64	0
Without sharpness	4 / 16	0.68	0
Without NIMA	1 / 16	0.94	0

shown a second time and were not used as updates. Four occasions were held out and were scored only after every training update. A pick that differed from the photograph Lumina displayed first was one step (12), one pass, with a separate weight vector for each person. People are not pooled.

On the two repeated occasions, 16 of 26 second answers matched that person’s first answer, against about 10 if the second answer ignored the first. Seven of the ten changes moved to a photograph Lumina had ranked lower. People also disagree with each other. Table VI places every kept photograph in the displayed order. Rank 1 scores 1 and last place scores 0, so chance is 0.50 at every length. The 234 answers score 0.53. Almost all of that gain is the top photograph (93 against 84.5). It is not uniform: two-photo questions carry it, three-photo questions are half the answers and sit at 0.48, and four-photo questions are only slightly above chance on the top photograph.

One learning pass does not beat leaving the weights alone. On the 52 held-out answers the original first photograph matches 26 and the first photograph after that person’s updates matches 24. Three people gain a match and three lose matches. The other seven keep the same count.

TABLE VI

Where the kept photograph sits in Lumina’s displayed order. The 0–1 score maps rank 1 to 1 and last place to 0. Chance is a uniform pick among the photographs on that question. Lower mean rank is closer to the photograph Lumina showed first. Thirteen finished sittings, 18 questions.

Photos	Picks	Mean rank		Score		Top-1	
		Lumina	Chance	Lumina	Chance	Lumina	Chance
2	65	1.43	1.50	0.57	0.50	37	32.5
3	117	2.04	2.00	0.48	0.50	41	39
4	52	2.25	2.50	0.58	0.50	15	13
All	234	1.92	1.97	0.53	0.50	93	84.5

Three conclusions follow. Preference is inconsistent and it varies: a fixed order cannot match both showings when a person changes, and the thirteen people do not share one taste, so their sittings stay separate. A score of 0.53 against 0.50 is already evidence that the signals are not noise. It is not a reason to keep a single best picture and delete the rest. A deletion has to show its reason, or a close call will not be trusted. The seven weights are also incomplete. On one three-photograph occasion every tester kept the frame Lumina ranked last: the wide portrait with the whole party, not the tighter views of three people that scored higher. That is sentimental value in capturing everybody who was there. Retuning sharpness or NIMA cannot add it.

VI. Discussion

Zero cannot-link merges do not mean the cast is correct. There are no identity labels, so Section V-B says nothing about the same person split apart, and a high α can still trust a wrong ArcFace vector. The event result is one distance matrix and one person’s grouping of 20 occasions. Twenty-three photographs have no EXIF time, and time cannot split identical scenes once the threshold exceeds 0.35. The ranker still has two internal mismatches: per-person face signals, other than eyes, drop the floors of

Table II, and MMR can open on a blink because it reads S rather than the constrained order. The retouch guard misses a changed background or a face the detector never saw.

Further work follows the keep-one study. Repeat more than two occasions, and keep one weight vector per person, so inconsistency is measured rather than averaged away. Stop treating one best picture as the curation: several frames can remain, and a frame set aside should show which signal caused it. The application’s Cleanup view is a first step: it labels each frame it sets aside as a duplicate, a blink or a blur, and deletes only when the user confirms. Extend the seven weights. The first candidate is how many of the occasion’s people are in the frame, which is the group-portrait case above, not a larger NIMA weight. A public album with identity labels, a second grouping of the occasions, and Recall@ K for CLIP search would strengthen the other claims. They do not replace that measurement.

VII. Conclusion

On this collection the fused clustering makes no cannot-link merge, the label-free event threshold matches the best labelled threshold, and the blink constraint is what keeps closed eyes out of the winners. Thirteen people put the displayed order only slightly above a random pick. Their preferences vary, a single keeper is too aggressive unless the deletion says why, and the seven weights do not yet count the value of having everybody in the frame.

Acknowledgment

Ong Qing Quan integrated the baseline pipeline and the web application, and carried the Semester 2 model improvements and the side models. Agustinus Benjamin Prasetyo designed the original Semester 1 model, the evaluation and the human study. An AI coding assistant was used to implement parts of the software and to draft this paper from the codebase, the Semester 1 report and the experiment logs. The authors remain responsible for the text.

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